Improve Multi-OS Computer Performance

...through Cross Swapping

Computers with more than one hard disk and more than one operating system (OS) can be set up to provide better performance than what OS installations generally default to. This technique, called cross swapping by the author, is described using a dual-boot scenario of Linux and Microsoft Windows.

Let's first understand what virtual memory is. Today, computers are used for a wide variety of tasks, facilitated in part by modern OSs providing multitasking capabilities. A multitasking OS allows you to run more than one program at a time. If the number of programs you have started is greater than the number of CPUs (processors) available and in use, the OS repeatedly runs tiny parts of each program for very small amounts of time (not noticeable by the user) by turn; that is, it repeatedly allows each program to use the available CPU(s) for tiny fractions of a second. This creates the illusion of all programs running in parallel.

Multitasking, however, places large demands on the system memory (RAM). Running several graphical programs simultaneously can easily exhaust even today's large memory banks in a short while. To alleviate this problem, modern OSs move some unused program data from RAM onto the hard disk to make room for required data.

This can be done because all of a program's data in RAM is not needed at all times. If a program has finished working with some data and needs other data that had earlier been moved out to the hard disk, the OS can again move out the used data and move the old data back in. This entire process of moving unneeded data out and required data in is called swapping or paging, and the area on the hard disk where data is moved out from the RAM is called the swap space, page file, or virtual memory. Virtual memory is used even if you do not run more than one program at a time, simply because